

CODEHIVE

JAVA PROGRAMMING: LOGIC & BOOLEANS GUIDE

► What is a Boolean?

In Java, the **boolean** data type allows the program to make decisions. It represents a logic gate that can only be either **true** or **false**.

```
boolean isJavaFun = true;
boolean isFishTasty = false;

System.out.println(isJavaFun);    // Result: true
```

► Logic Expressions

Expressions that use comparison operators return a boolean value.

```
System.out.println(10 > 9);    // true
System.out.println(5 == 10);  // false
```

- **==** Equal to | **!=** Not equal
- **>** / **<** Greater / Less than

► Decision Making: If-Else

Booleans are the engines behind `if` statements.

```
int myAge = 25;
int votingAge = 18;

if (myAge >= votingAge) {
    System.out.println("Old enough to vote!");
} else {
    System.out.println("Access Denied.");
}
```

► Logical Operators

Connecting multiple boolean conditions into one.

- **&& (AND)**: True if both sides are true.
- **|| (OR)**: True if at least one side is true.
- **! (NOT)**: Reverses the result (true becomes false).

```
boolean hasID = true;
boolean isAdult = true;

System.out.println(hasID && isAdult); // true
```

► CodeHive Logic Summary

- Booleans store only **true** or **false**.
- **Boolean Expressions** power all software decision-making.
- **Logical Operators** allow complex conditions (AND, OR, NOT).
- Default value for boolean class fields is **false**.